

Distributional Empirical Discipline: Data and Current Model

Built May 8, 2026

This note is a discipline file, not a validation claim. For each empirical object that is currently available, it shows the data object and the closest object in the active Python model. The model is the current direct-geometry benchmark, job 37, evaluation 1425, with the renewal-valve stationary closure. The benchmark loss in the cluster record is 8.128 and the resolved equilibrium error is 0.0030.

The organizing economic references are standard: Becker and Becker–Lewis for the fertility quantity-quality tradeoff; Sommer and Doepke–Kindermann style lifecycle fertility models for income and lifecycle discipline; Rosen–Roback, Henderson–Ioannides, Sommer–Sullivan–Verbrugge, and Gyourko–Mayer–Sinai for spatial sorting, tenure, and housing-price discipline. The point of this file is to make clear which of those cross-sectional facts the current model can speak to now.

Income needs special care. The model has a deterministic lifecycle income profile, so income moves with age. It does not have persistent idiosyncratic income risk or a within-age income distribution. Therefore the model income-bin plot below uses the stationary mass generated by deterministic age/location income; it is a lifecycle-rank diagnostic, not a clean analog of PSID young-adult income quintiles.

Table 1: Benchmark calibration moments, with targets next to model moments.

Moment	Target	Model
tfr	1.700	1.898
childless_rate	0.150	0.145
mean_age_first_birth	26.000	33.535
tfr_gradient	0.133	0.119
own_rate	0.627	0.643
own_gradient	0.170	0.139
own_family_gap	0.110	0.114
prime_childless_renter_median_rooms	4.000	6.365
prime_childless_owner_median_rooms	6.000	6.800
housing_increment_0to1	0.664	0.441
housing_increment_1to2	0.566	0.192
young_liquid_wealth_to_income	0.600	0.527
center_share_nonparents	0.494	0.405
center_share_newparents	0.416	0.382
migration_rate	0.032	0.035
old_age_own_rate	0.863	0.947
old_age_parent_childless_gap	0.070	0.062
inv_pop_share_C	0.450	0.441
inv_rent_ratio_C_over_P	1.140	1.182

Note: These are the live benchmark moments from the current direct-geometry calibration record. Distributional tables below are not additional calibration targets.

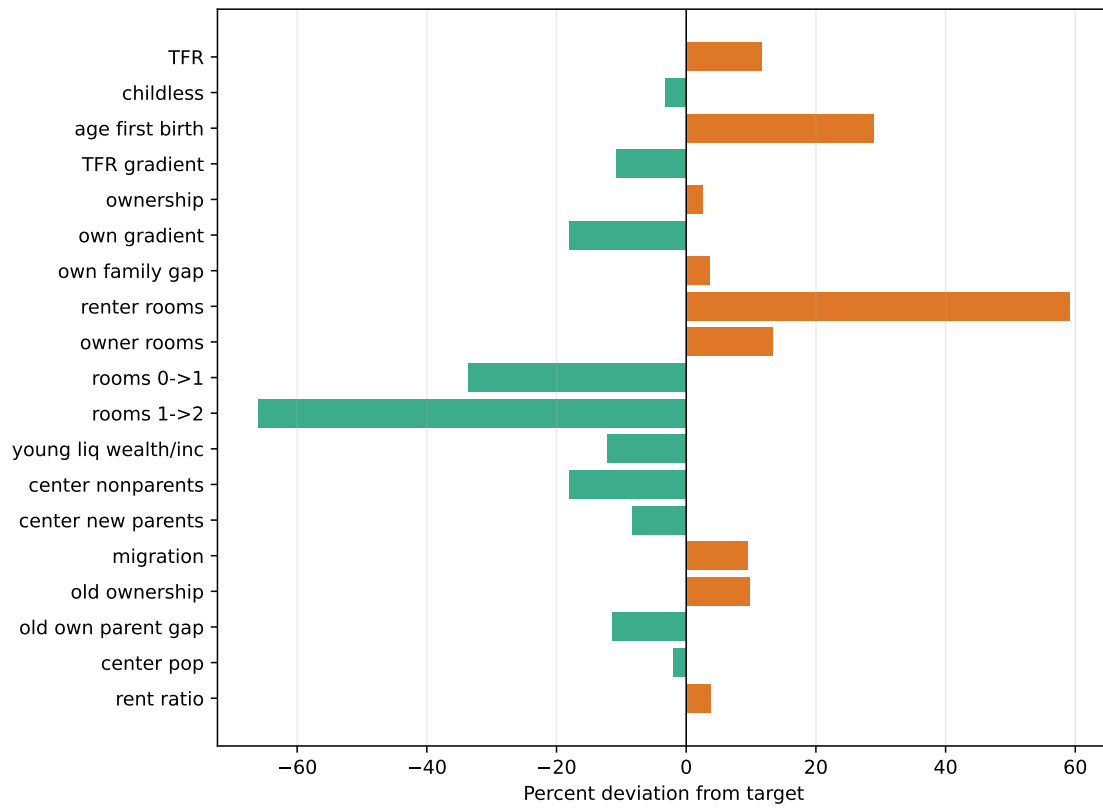


Figure 1: Calibration target deviations. Positive values mean the model is above the target.

Income

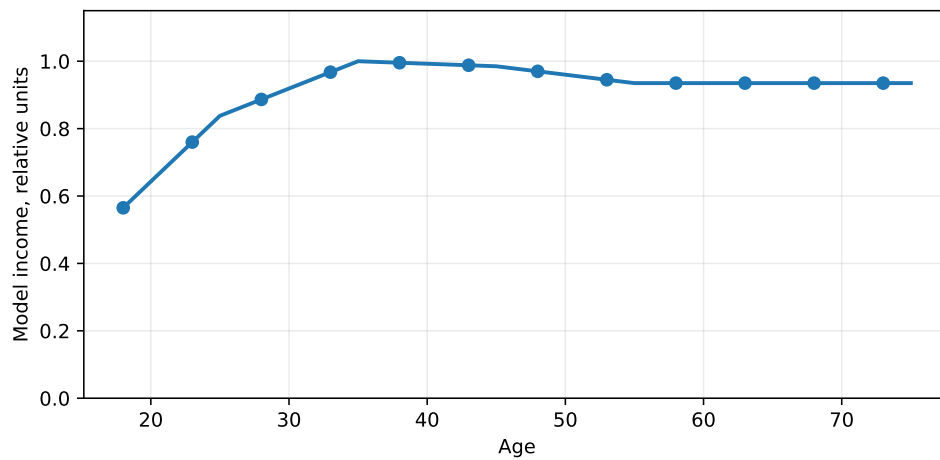


Figure 2: Model lifecycle income profile. This is the deterministic age component that creates income dynamics in the model.

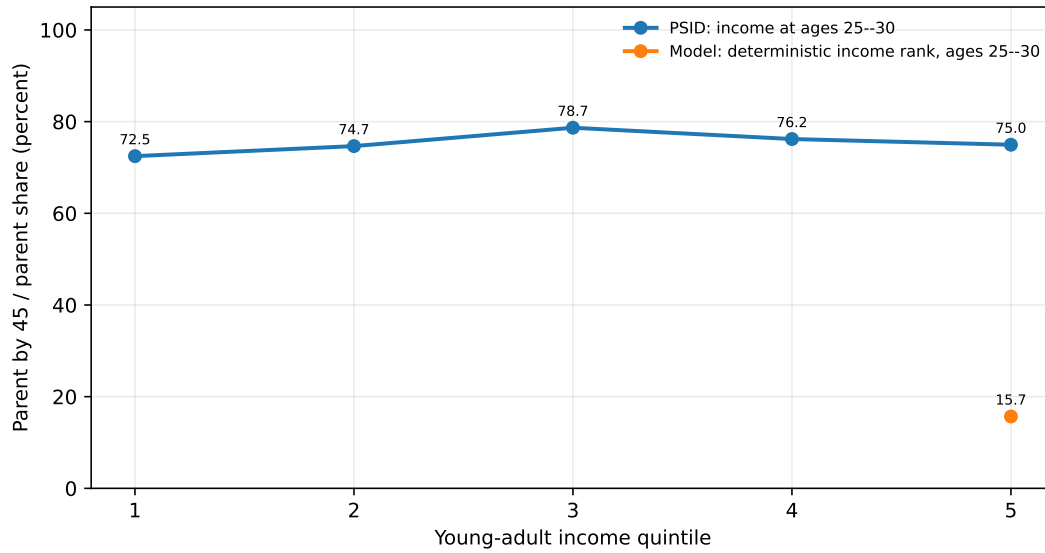


Figure 3: Parenthood by young-adult income bins. Data use PSID family income averaged over ages 25–30. The model uses ages 25–30 current income, which varies with age and location but not with idiosyncratic permanent income. Missing model points are empty rank bins.

Wealth And Tenure

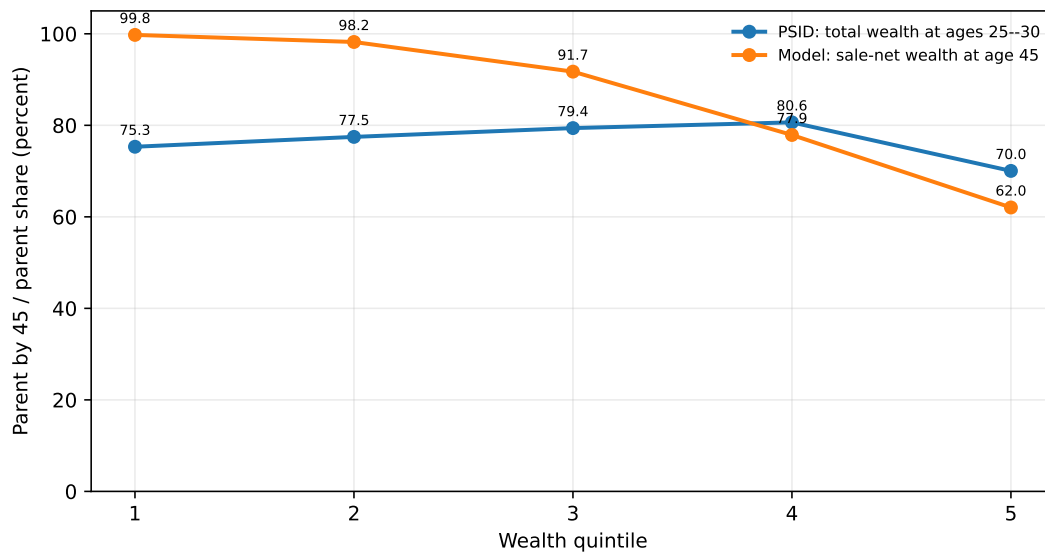


Figure 4: Parenthood by wealth bins. Data use young-adult PSID total wealth; model uses cross-sectional sale-net wealth at age 45.

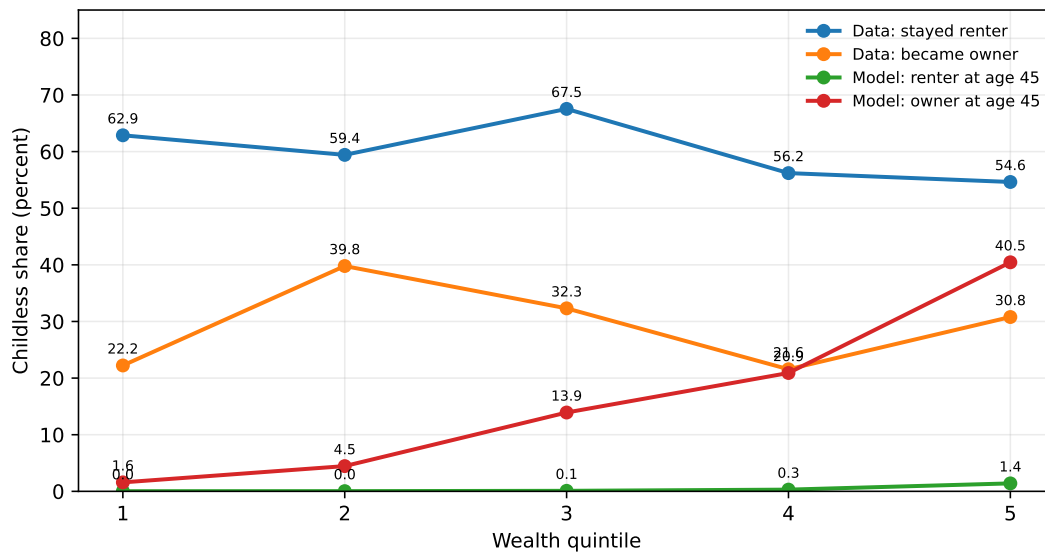


Figure 5: Childlessness by wealth and tenure. The data object follows young renters by whether they become owners by age 35; the model object is current tenure at age 45.

Housing Space

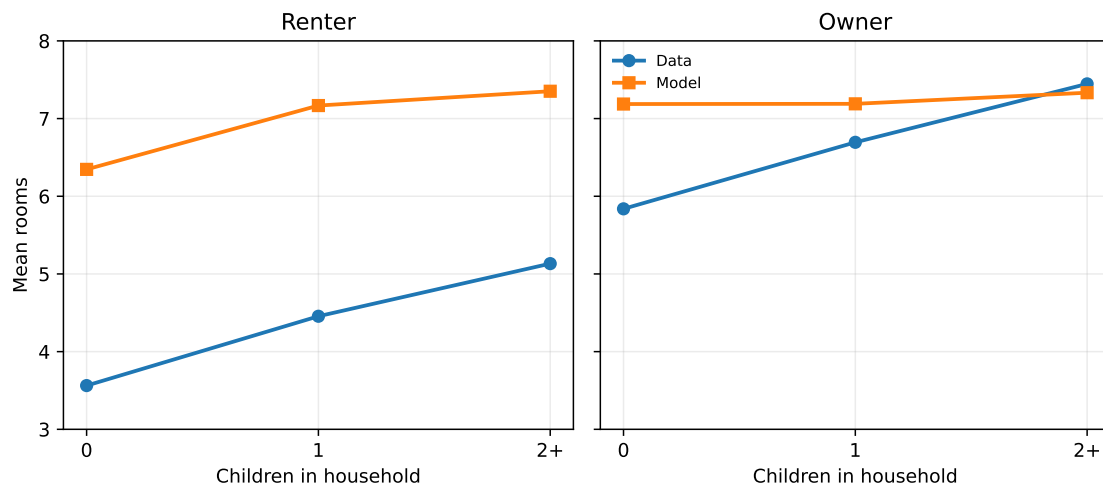


Figure 6: Mean rooms by tenure and children, ACS versus model.

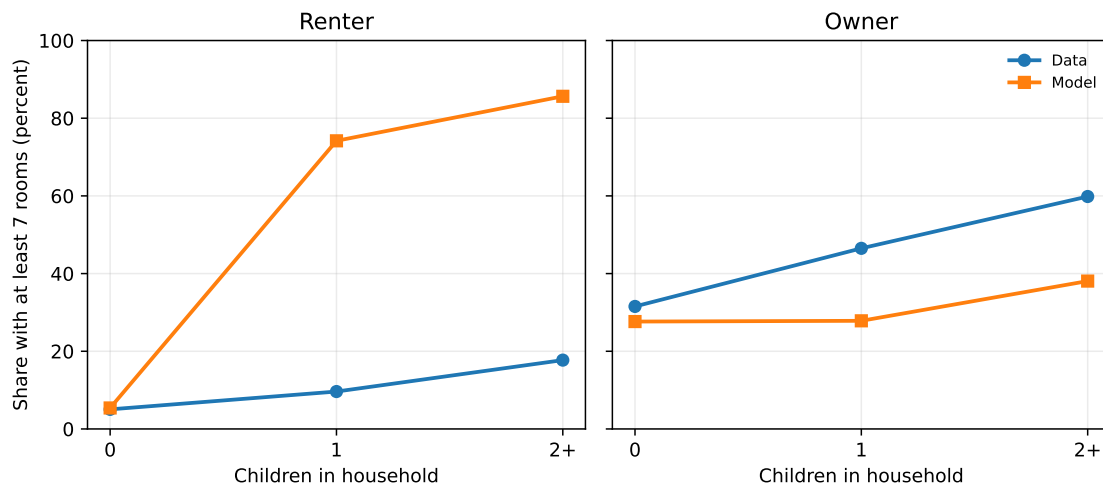


Figure 7: Share of households with at least 7 rooms by tenure and children, ACS versus model.

Space And Mobility

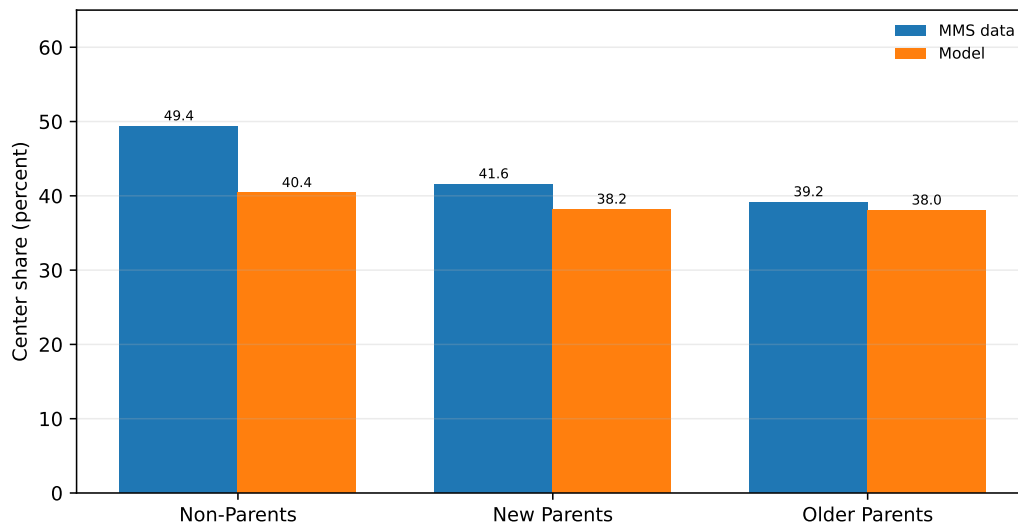


Figure 8: Center residence shares by parent status, MMS data versus model.

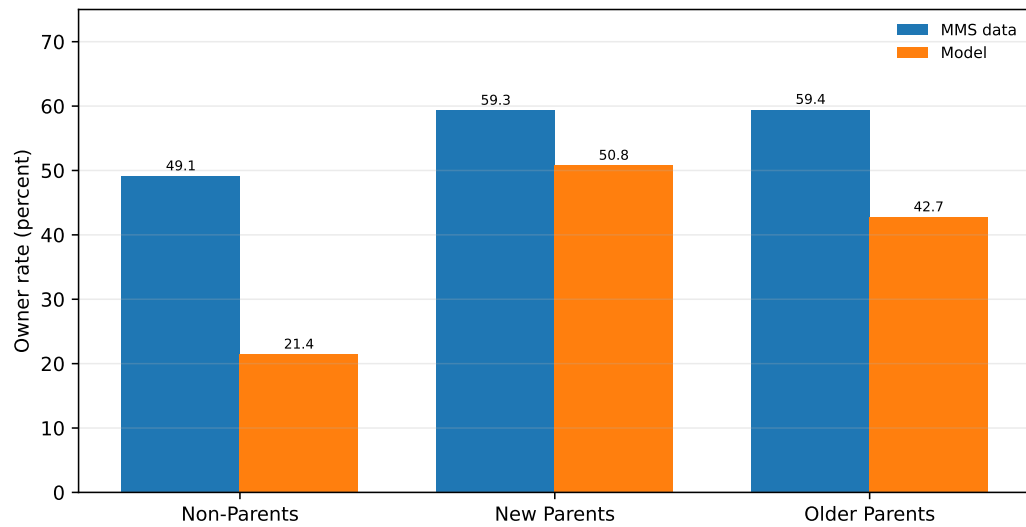


Figure 9: Owner rates by parent status, MMS data versus model.

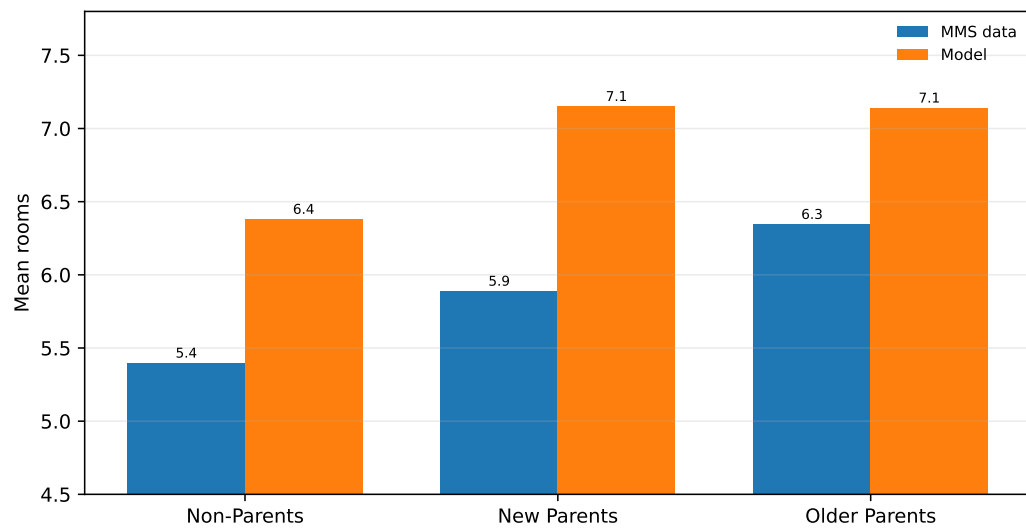


Figure 10: Mean rooms by parent status, MMS data versus model.

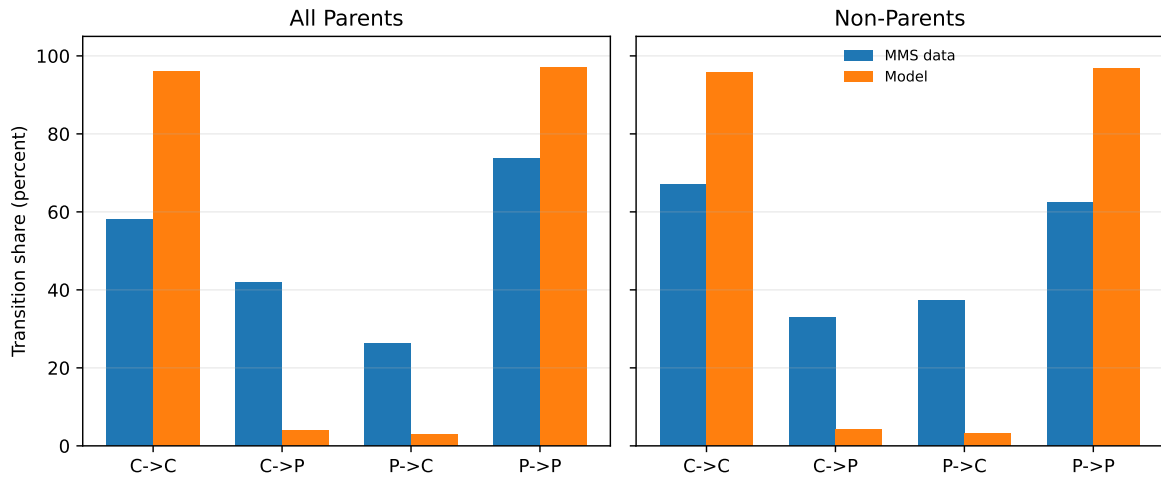


Figure 11: Center/periphery origin-destination shares. The model analog is the one-period location-choice probability averaged over the stationary distribution.

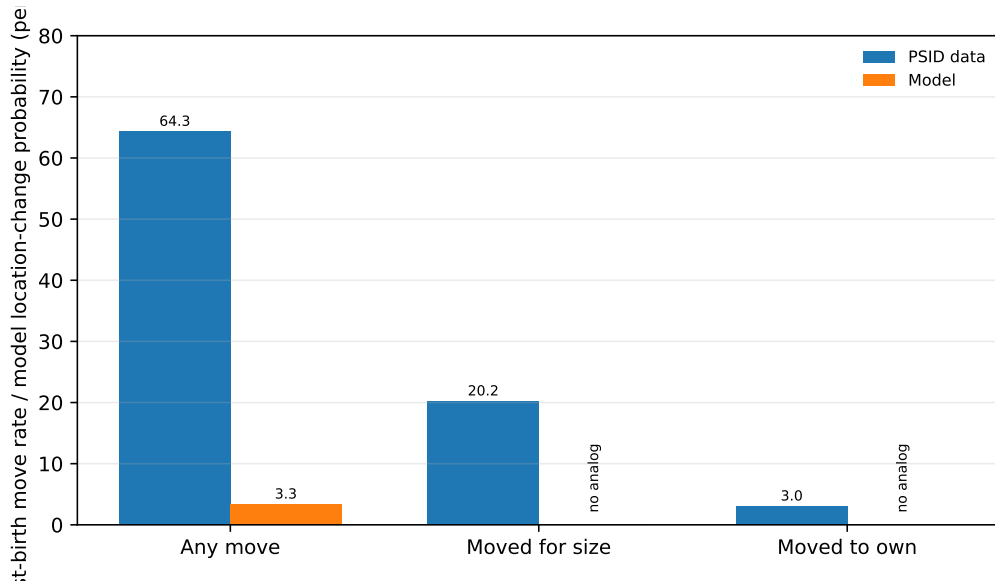


Figure 12: First-birth moving facts. The current model has an analog for any location change, but no move-for-size or moved-to-ownership reason code.

Read

The model has genuine lifecycle income dynamics, but not the cross-sectional income heterogeneity needed to match PSID income quintile fertility facts. The wealth comparison is closer, but still not exact because the data object is young-adult wealth predicting later fertility while the model object is a stationary cross-section. The room and spatial plots are the cleanest diagnostics: renter housing services are too high, the model does not generate enough room dispersion by tenure and children, and center/periphery transition rates are much too low. These are useful failures because they identify exactly which empirical margins the current model cannot yet claim to validate.